



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Solve 2×2 systems by substitution and elimination
- Solve compound and absolute value inequalities
- Interpret the solution of a system in context
- Graph solution sets of systems on the number line

Simplify each expression completely. Show all steps and circle your final answer.

Compound inequalities

1. Solve the compound inequality: $7 < 2x + 1 < 21$

$$7 < 2x + 1 < 21$$

Answer: _____

2. A temperature T (in Celsius) satisfies $7 < 2T + 4 < 30$. Solve for T .

$$7 < 2x + 4 < 30$$

Answer: _____

3. Solve the compound inequality: $6 < 3x + 3 < 21$

$$6 < 3x + 3 < 21$$

Answer: _____

4. A temperature T (in Celsius) satisfies $7 < 3T + 2 < 26$. Solve for T .

$$7 < 3x + 2 < 26$$

Answer: _____

5. Solve the compound inequality: $5 < 4x + 5 < 22$

$$5 < 4x + 5 < 22$$

Answer: _____

6. A temperature T (in Celsius) satisfies $6 < 3T + 3 < 21$. Solve for T .

$$6 < 3x + 3 < 21$$

Answer: _____



7. Solve the compound inequality: $5 < 2x + 2 < 16$

$$5 < 2x + 2 < 16$$

Answer: _____

8. A temperature T (in Celsius) satisfies $7 < 2T + 4 < 25$. Solve for T .

$$7 < 2x + 4 < 25$$

Answer: _____

Linear inequalities

9. Solve and write the solution: $4x + 9 > 23$. Graph the solution on a number line.

$$4x + 9 > 23$$

Answer: _____

10. Solve and express the solution: $3x + 7 < 23$.

$$3x + 7 < 23$$

Answer: _____

11. Solve and write the solution: $1x + 7 \leq 7$. Graph the solution on a number line.

$$1x + 7 \leq 7$$

Answer: _____

12. Solve: $4x - 2 \geq 16$.

$$4x - 2 \geq 16$$

Answer: _____

13. Solve and write the solution: $5x + 4 > 20$. Graph the solution on a number line.

$$5x + 4 > 20$$

Answer: _____

14. Solve and express the solution: $2x + 8 < 25$.

$$2x + 8 < 25$$

Answer: _____

15. Solve and write the solution: $1x + 4 \leq 10$. Graph the solution on a number line.

$$1x + 4 \leq 10$$

Answer: _____



16. Solve: $6x - 1 \geq 18$.

$$6x - 1 \geq 18$$

Answer: _____

17. Solve and write the solution: $5x + 3 > 8$. Graph the solution on a number line.

$$5x + 3 > 8$$

Answer: _____

18. Solve and express the solution: $4x + 8 < 17$.

$$4x + 8 < 17$$

Answer: _____

19. Solve and write the solution: $4x + 2 \leq 26$. Graph the solution on a number line.

$$4x + 2 \leq 26$$

Answer: _____

20. Solve: $5x - 5 \geq 12$.

$$5x - 5 \geq 12$$

Answer: _____

21. Solve and write the solution: $3x + 9 > 25$. Graph the solution on a number line.

$$3x + 9 > 25$$

Answer: _____

Systems of equations

22. Solve the system by elimination. Add the two equations to eliminate y.

$$\begin{cases} 3x - 2y = 0 \\ 3x + 2y = 12 \end{cases}$$

Answer: _____

23. Solve the system by elimination: $1x + 3y = 8$ $4x + 1y = -1$

$$\begin{cases} x + 3y = 8 \\ 4x + y = -1 \end{cases}$$

Answer: _____



24. Solve the system by substitution.

$$\begin{cases} y = 0x + 0 \\ 3x + 2y = 6 \end{cases}$$

Answer: _____

25. Solve the system by elimination. Add the two equations to eliminate y.

$$\begin{cases} x - 3y = 4 \\ x + 2y = -1 \end{cases}$$

Answer: _____

26. Solve the system by elimination: $2x + 2y = -4$ $2x + 1y = -4$

$$\begin{cases} 2x + 2y = -4 \\ 2x + y = -4 \end{cases}$$

Answer: _____

27. Solve the system by substitution.

$$\begin{cases} y = -2x - 3 \\ 4x + y = -5 \end{cases}$$

Answer: _____

28. Solve the system by elimination. Add the two equations to eliminate y.

$$\begin{cases} x - 3y = 3 \\ 4x + 2y = 12 \end{cases}$$

Answer: _____

29. Solve the system by elimination: $1x + 1y = -1$ $4x + 4y = -4$

$$\begin{cases} x + y = -1 \\ 4x + 4y = -4 \end{cases}$$

Answer: _____

30. Solve the system by substitution.

$$\begin{cases} y = 2x + 3 \\ x + 2y = 0 \end{cases}$$

Answer: _____





Topics: Compound inequalities, Linear inequalities, Systems of equations. All answers verified by independent computation.

Solutions



Compound inequalities

1. Solve the compound inequality: $7 < 2x + 1 < 21$

$$7 < 2x + 1 < 21$$

→ Subtract 1 from all three parts: $7 - 1 < 2x < 21 - 1$.

→ Divide all three parts by 2.

→ Solution: $3 < x < 10$.

Answer: $3 < x < 10$

2. A temperature T (in Celsius) satisfies $7 < 2T + 4 < 30$. Solve for T .

$$7 < 2x + 4 < 30$$

→ Subtract 4 throughout: $7 - 4 < 2T < 30 - 4$.

→ Divide by 2: solution is $3/2 < x < 13$.

→ The temperature T falls in this interval.

Answer: $\frac{3}{2} < x < 13$

3. Solve the compound inequality: $6 < 3x + 3 < 21$

$$6 < 3x + 3 < 21$$

→ Subtract 3 from all three parts: $6 - 3 < 3x < 21 - 3$.

→ Divide all three parts by 3.

→ Solution: $1 < x < 6$.

Answer: $1 < x < 6$

4. A temperature T (in Celsius) satisfies $7 < 3T + 2 < 26$. Solve for T .

$$7 < 3x + 2 < 26$$

→ Subtract 2 throughout: $7 - 2 < 3T < 26 - 2$.

→ Divide by 3: solution is $5/3 < x < 8$.

→ The temperature T falls in this interval.

Answer: $\frac{5}{3} < x < 8$

5. Solve the compound inequality: $5 < 4x + 5 < 22$

$$5 < 4x + 5 < 22$$

→ Subtract 5 from all three parts: $5 - 5 < 4x < 22 - 5$.

→ Divide all three parts by 4.

→ Solution: $0 < x < 17/4$.

Answer: $0 < x < \frac{17}{4}$



6. A temperature T (in Celsius) satisfies $6 < 3T + 3 < 21$. Solve for T .

$$6 < 3x + 3 < 21$$

→ Subtract 3 throughout: $6 - 3 < 3T < 21 - 3$.

→ Divide by 3: solution is $1 < x < 6$.

→ The temperature T falls in this interval.

Answer: $1 < x < 6$

7. Solve the compound inequality: $5 < 2x + 2 < 16$

$$5 < 2x + 2 < 16$$

→ Subtract 2 from all three parts: $5 - 2 < 2x < 16 - 2$.

→ Divide all three parts by 2.

→ Solution: $3/2 < x < 7$.

Answer: $\frac{3}{2} < x < 7$

8. A temperature T (in Celsius) satisfies $7 < 2T + 4 < 25$. Solve for T .

$$7 < 2x + 4 < 25$$

→ Subtract 4 throughout: $7 - 4 < 2T < 25 - 4$.

→ Divide by 2: solution is $3/2 < x < 21/2$.

→ The temperature T falls in this interval.

Answer: $\frac{3}{2} < x < \frac{21}{2}$



Linear inequalities

9. Solve and write the solution: $4x + 9 > 23$. Graph the solution on a number line.

$$4x + 9 > 23$$

→ Subtract 9 from both sides: $4x > 14$.

→ Divide both sides by 4: $x > 7/2$.

→ Open circle at $7/2$, shade to the right.

Answer: $x > \frac{7}{2}$

10. Solve and express the solution: $3x + 7 < 23$.

$$3x + 7 < 23$$

→ Subtract 7: $3x < 16$. Divide by 3: $x < 16/3$.

Answer: $x < \frac{16}{3}$

11. Solve and write the solution: $1x + 7 \leq 7$. Graph the solution on a number line.

$$1x + 7 \leq 7$$

→ Subtract 7 from both sides: $1x \leq 0$.

→ Divide both sides by 1: $x \leq 0$.

→ Closed circle at 0, shade to the left.

Answer: $x \leq 0$

12. Solve: $4x - 2 \geq 16$.

$$4x - 2 \geq 16$$

→ Add 2: $4x \geq 18$. Divide by 4: $x \geq 9/2$.

Answer: $x \geq \frac{9}{2}$

13. Solve and write the solution: $5x + 4 > 20$. Graph the solution on a number line.

$$5x + 4 > 20$$

→ Subtract 4 from both sides: $5x > 16$.

→ Divide both sides by 5: $x > 16/5$.

→ Open circle at $16/5$, shade to the right.

Answer: $x > \frac{16}{5}$

14. Solve and express the solution: $2x + 8 < 25$.

$$2x + 8 < 25$$

→ Subtract 8: $2x < 17$. Divide by 2: $x < 17/2$.

Answer: $x < \frac{17}{2}$



15. Solve and write the solution: $1x + 4 \leq 10$. Graph the solution on a number line.

$$1x + 4 \leq 10$$

→ Subtract 4 from both sides: $1x \leq 6$.

→ Divide both sides by 1: $x \leq 6$.

→ Closed circle at 6, shade to the left.

Answer: $x \leq 6$

16. Solve: $6x - 1 \geq 18$.

$$6x - 1 \geq 18$$

→ Add 1: $6x \geq 19$. Divide by 6: $x \geq 19/6$.

Answer: $x \geq \frac{19}{6}$

17. Solve and write the solution: $5x + 3 > 8$. Graph the solution on a number line.

$$5x + 3 > 8$$

→ Subtract 3 from both sides: $5x > 5$.

→ Divide both sides by 5: $x > 1$.

→ Open circle at 1, shade to the right.

Answer: $x > 1$

18. Solve and express the solution: $4x + 8 < 17$.

$$4x + 8 < 17$$

→ Subtract 8: $4x < 9$. Divide by 4: $x < 9/4$.

Answer: $x < \frac{9}{4}$

19. Solve and write the solution: $4x + 2 \leq 26$. Graph the solution on a number line.

$$4x + 2 \leq 26$$

→ Subtract 2 from both sides: $4x \leq 24$.

→ Divide both sides by 4: $x \leq 6$.

→ Closed circle at 6, shade to the left.

Answer: $x \leq 6$

20. Solve: $5x - 5 \geq 12$.

$$5x - 5 \geq 12$$

→ Add 5: $5x \geq 17$. Divide by 5: $x \geq 17/5$.

Answer: $x \geq \frac{17}{5}$



21. Solve and write the solution: $3x + 9 > 25$. Graph the solution on a number line.

$$3x + 9 > 25$$

→ Subtract 9 from both sides: $3x > 16$.

→ Divide both sides by 3: $x > 16/3$.

→ Open circle at $16/3$, shade to the right.

Answer: $x > \frac{16}{3}$



Systems of equations

22. Solve the system by elimination. Add the two equations to eliminate y .

$$\begin{cases} 3x - 2y = 0 \\ 3x + 2y = 12 \end{cases}$$

→ Notice the y -terms have opposite signs. Add the equations to eliminate y .

→ $(3+3)x = 12$, so $x = 12/6$.

→ Substitute x back to find y .

→ Solution: $x = 2$, $y = 3$.

Answer: $x = 2$, $y = 3$

23. Solve the system by elimination: $1x + 3y = 8$ $4x + 1y = -1$

$$\begin{cases} x + 3y = 8 \\ 4x + y = -1 \end{cases}$$

→ Multiply and eliminate one variable.

→ $x = -1$, $y = 3$.

Answer: $x = -1$, $y = 3$

24. Solve the system by substitution.

$$\begin{cases} y = 0x + 0 \\ 3x + 2y = 6 \end{cases}$$

→ The first equation gives y in terms of x directly.

→ Substitute into the second equation and solve for x .

→ Back-substitute to find y .

→ Solution: $x = 2$, $y = 0$.

Answer: $x = 2$, $y = 0$

25. Solve the system by elimination. Add the two equations to eliminate y .

$$\begin{cases} x - 3y = 4 \\ x + 2y = -1 \end{cases}$$

→ Notice the y -terms have opposite signs. Add the equations to eliminate y .

→ $(1+1)x = 3$, so $x = 3/2$.

→ Substitute x back to find y .

→ Solution: $x = 1$, $y = -1$.

Answer: $x = 1$, $y = -1$

26. Solve the system by elimination: $2x + 2y = -4$ $2x + 1y = -4$

$$\begin{cases} 2x + 2y = -4 \\ 2x + y = -4 \end{cases}$$

→ Multiply and eliminate one variable.

→ $x = -2$, $y = 0$.

Answer: $x = -2$, $y = 0$



27. Solve the system by substitution.

$$\begin{cases} y = -2x - 3 \\ 4x + y = -5 \end{cases}$$

→ The first equation gives y in terms of x directly.
→ Substitute into the second equation and solve for x .
→ Back-substitute to find y .
→ Solution: $x = -2$, $y = 3$.

Answer: $x = -2$, $y = 3$

28. Solve the system by elimination. Add the two equations to eliminate y .

$$\begin{cases} x - 3y = 3 \\ 4x + 2y = 12 \end{cases}$$

→ Notice the y -terms have opposite signs. Add the equations to eliminate y .
→ $(1+4)x = 15$, so $x = 15/5$.
→ Substitute x back to find y .
→ Solution: $x = 3$, $y = 0$.

Answer: $x = 3$, $y = 0$

29. Solve the system by elimination: $1x + 1y = -1$ $4x + 4y = -4$

$$\begin{cases} x + y = -1 \\ 4x + 4y = -4 \end{cases}$$

→ Multiply and eliminate one variable.
→ $x = 0$, $y = -1$.

Answer: $x = 0$, $y = -1$

30. Solve the system by substitution.

$$\begin{cases} y = 2x + 3 \\ x + 2y = 0 \end{cases}$$

→ The first equation gives y in terms of x directly.
→ Substitute into the second equation and solve for x .
→ Back-substitute to find y .
→ Solution: $x = 0$, $y = 0$.

Answer: $x = 0$, $y = 0$

